Simran Mann

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EDUCATION

Simon Fraser University Sept 2022 – Dec 2026
Bachelor of Applied Science in Computing Science GPA: 3.99/4.33

- **Relevant Courses**: Distributed Systems, Compilers, Computer Architecture, Operating Systems

TECHNICAL SKILLS

Programming Languages: C/C++ (multithreading, atomics, STL, RAII), Python, x86-64 Assembly, Bash
Systems & Concurrency: POSIX threads, std::thread, mutexes, atomics, MPI, lock-free progress guarantees
Embedded Systems & Firmware: STM32 & ATSAME MCUs, SPI, I2C, UART, GPIO, DMA
Tools & Platforms: Linux/Unix, Git, CMake, GTest, Docker, LLVM, STM32

EXPERIENCE

SFU Robot Soccer May 2025 – Present
Firmware Developer

- Develop firmware for autonomous soccer robots on ATSAME70 MCUs as part of a 6-person team, implementing BLDC motor control, SPI communication for SX1276 LoRa and IMU, and real-time command execution
- Designed firmware for a basestation to parse protobuf commands over UART and relay them wirelessly via SX1276 LoRa with CRC error detection, extracting functionality from datasheets to configure LoRa parameters
- Implemented movement functions by translating forward/left speed commands into PWM duty cycles for each BLDC motor, coordinating translation, strafing, diagonal, and rotation movement types
- Collaborate with software team on CI/CD integration to ensure robots accurately execute camera-guided navigation commands

NETGEAR Sep 2024 – April 2025
Orbi Software Test Engineer

- Designed and deployed multi-node network testbeds using managed switches, configuring VLANs, static routing, and network topologies to simulate mesh network environments
- Developed Python automation tool to parse UART serial output, configure devices remotely, and automate network stability testing, reducing test execution time from 8 hours to 2 hours
- Debugged network-level issues including packet loss, routing failures, and inter-device communication problems using Wireshark, tcpdump, ping, traceroute, and iperf for bandwidth analysis
- Validated router and mesh network behavior over TCP/UDP traffic, testing packet forwarding, protocol-level correctness, latency, and failover scenarios

PROJECTS

Parallel and Distributed Graph Algorithms Sept – Dec 2024

- Implemented parallel Google PageRank and Triangle Counting algorithms in C++ using std::thread for shared-memory systems and MPI for distributed multi-node clusters
- Scaled implementations from single-node multithreaded execution to distributed MPI-based execution, optimizing message-passing to reduce synchronization overhead
- Evaluated vertex-based, edge-based, and dynamic task decomposition approaches to improve load balance and inter-process communication

Concurrent Data Structures Oct – Nov 2024

- Implemented non-blocking queues, stacks, and sets using lock-free linked lists in C++
- Benchmarked data structures under multi-producer/multi-consumer workloads to analyze contention bottlenecks
- Addressed ABA problems via version-tagged pointers and compare-and-swap (CAS) atomic operations

C Language Compiler June – Aug 2024

- Developed compiler in C++ using Flex/Bison and LLVM APIs, implementing SSA-form IR code generation and short-circuit evaluation for booleans
- Built a complete pipeline from parsing through semantic analysis to code generation on linux
- Built C runtime and Python scripts to automate testing, validating compilation and execution of code